

Basic Electrical Engineering (BT EE103A)



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Unit-II

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AC Fundamental

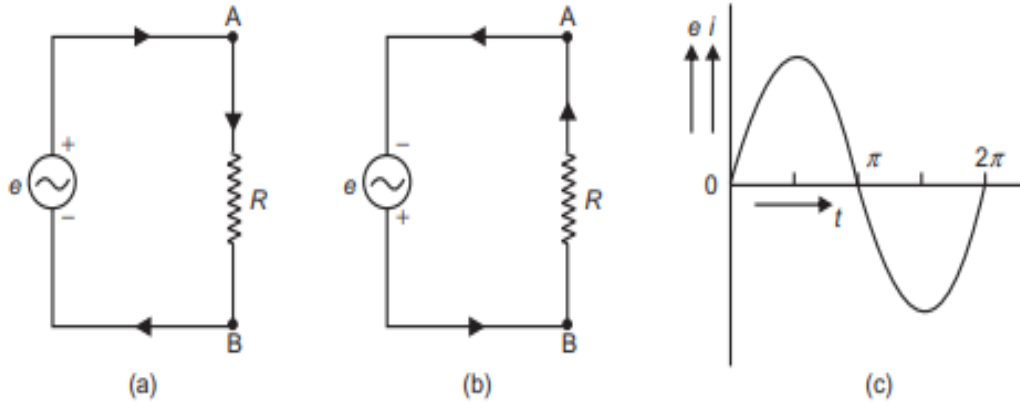


Fig 1. Alternating voltage and current (a) AC voltage applied across resistor, flow of current during first half cycle (b) AC voltage applied across resistor, flow of current during next half cycle

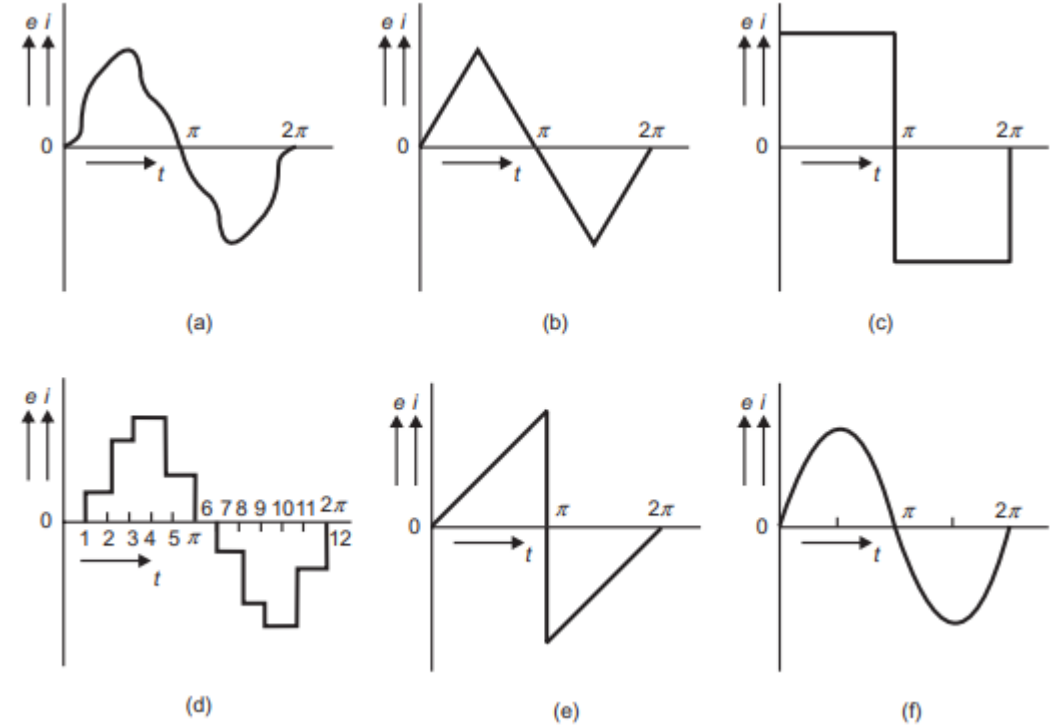


Fig. 2 AC wave shapes (a) General ac wave (b) Triangular wave (c) Square wave (d) Periodic wave (e) Triangular/saw tooth wave (f) Sinusoidal wave

Difference between ac and dc

AC	DC
(1) An alternating current reverses periodically and its magnitude changes.	(1) Direct current flows only in one direction and remains unaltered.
(2) Amplitude and polarities are varying continuously.	(2) Amplitude and polarities are fixed.
(3) It has a particular frequency.	(3) It is independent of frequency.
(4) AC can be generated at higher voltages.	(4) DC cannot be generated at high voltages because of commutation difficulties.
(5) In case of AC, the cost of generation is less.	(5) In case of DC, the cost of generation is more.
(6) Alternating voltage can be increased (stepped up or decreased (stepped down) easily with the help of a transformer.	(6) Direct voltage cannot be increased or decreased easily.
(7) AC motors are of less cost, more robust, and durable.	(7) DC motors are costly and less durable.
(8) The maintenance cost of AC equipment and appliances is less.	(8) The maintenance cost of DC equipment and appliances is more.
(9) AC cannot be used directly for electroplating.	(9) Only DC can be used directly for electroplating.
(10) The speed of AC motors cannot be controlled easily.	(10) The speed control of DC motor is very easy and economical.

Sinusoidal alternating quantity

An alternating quantity (i.e., voltage or current) that varies according to sine of angle q ($q = \omega t$) is known as 'sinusoidal alternating quantity'. Its wave shape is shown in Figure 2(f). For the generation of electric power, sinusoidal voltages and currents are selected all over the world due to the following reasons:

1. The sinusoidal voltages and currents cause low iron and copper losses in AC rotating machines and transformers. This improves the efficiency of AC machines.
2. The sinusoidal voltages and currents offer less interference to nearby communication system (e.g., telephone lines)
3. They produce least disturbance in the electrical circuits.
- 4.

Whenever the word 'alternating voltage or current' is used in this text, it means sinusoidal alternating voltage or current unless stated otherwise

Generation of alternating voltage and current

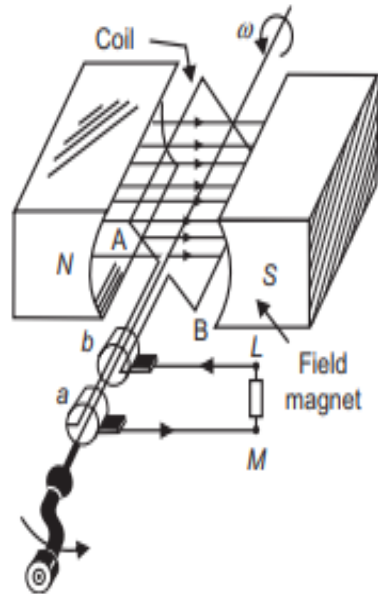


Fig. 3 Production of ac voltage (rotating coil, field stationary)

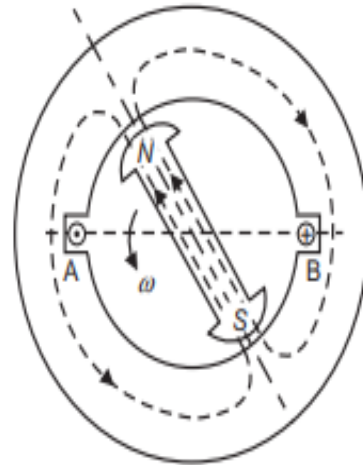


Fig. 4 Production of ac voltage (rotating field, coil stationary)

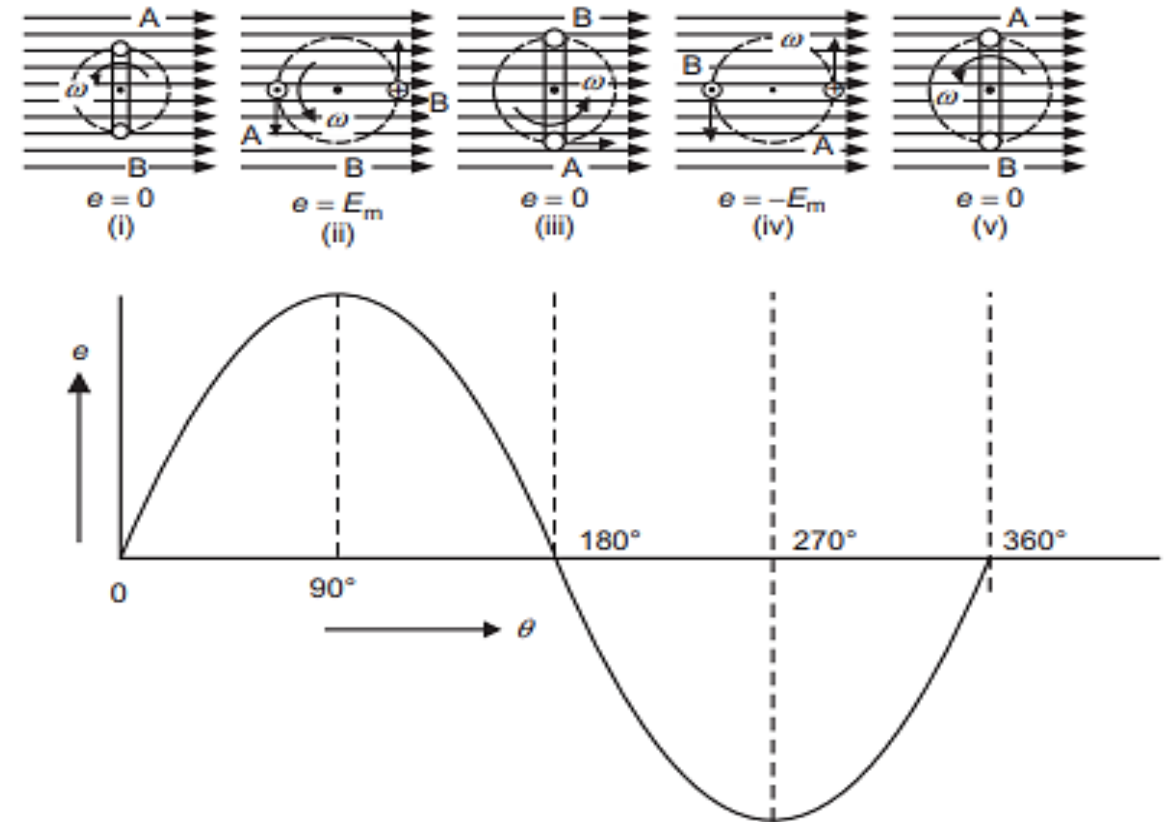


Fig. 5 Position of coil at various instants. Wave shape of generated voltage

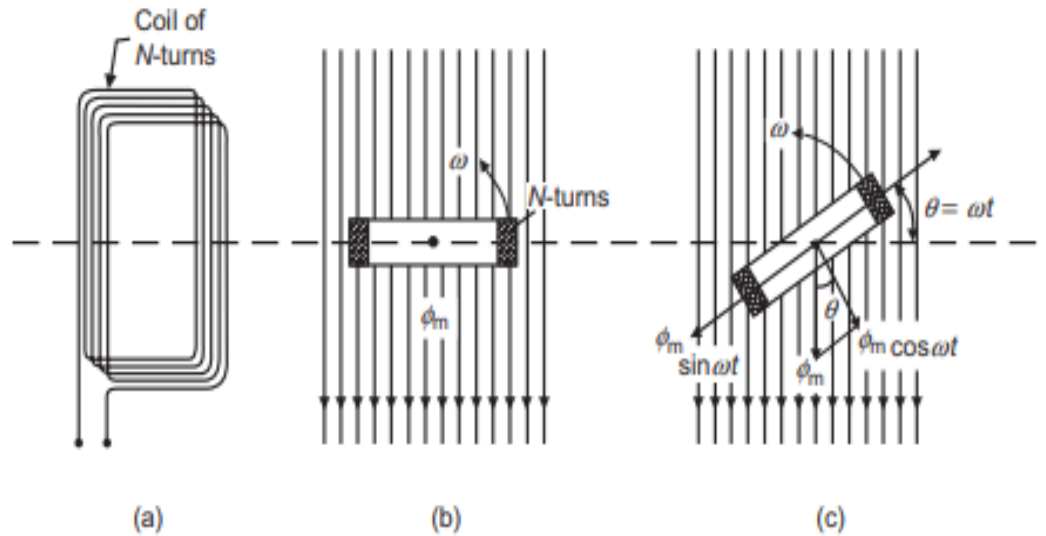


Fig. 6 Multi-turn coil rotating in a constant magnetic field (a) Multi-turn coil (b) Position of coil at an instant (c) Position of coil after t second

According to Faraday's laws of electromagnetic induction, the magnitude of emf induced in the coil at this instant, that is,
Instantaneous value of emf induced in the coil,

$e = -N \frac{d\phi}{dt}$ (negative sign indicates that an induced emf is opposite to the very cause which produces it)

$$\text{or } e = -N \frac{d}{dt} \phi_m \cos \omega t \quad (\phi = \phi_m \cos \omega t)$$

$$\text{or } e = -N \phi_m (-\omega \sin \omega t)$$

$$\text{or } e = \omega N \phi_m \sin \omega t$$

The value of an induced emf will be maximum when angle θ

$$\text{or } \omega t = 90^\circ \text{ (i.e., } \sin \omega t = 1)$$

$$\therefore E_m = \omega N \phi_m$$

$$e = E_m \sin \omega t = E_m \sin \theta$$

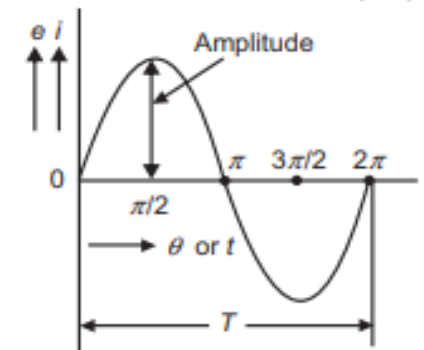


Fig. 7 Wave shape of a sinusoidal voltage or current

1. **Wave form:** The shape of the curve obtained by plotting the instantaneous values of alternating quantity (voltage or current) along Y-axis and time or angle ($q = \omega t$) along X-axis is called 'wave form or wave shape'.
2. **Instantaneous value:** The value of an alternating quantity, that is, voltage or current at any instant is called its instantaneous value and is represented by 'e' or 'i', respectively.
3. **Cycle:** When an alternating quantity goes through a complete set of positive and negative values or goes through 360 electrical degrees, it is said to have completed one cycle.
4. **Alternation:** One half-cycle is called 'alternation'. An alternation spans 180 electrical degrees.
5. **Time period:** The time taken in seconds to complete one cycle by an alternating quantity is called time period. It is generally denoted by 'T'.
6. **Frequency:** The number of cycles made per second by an alternating quantity is called 'frequency'. It is measured in cycles per second (c/s) or hertz (Hz) and is denoted by 'f'.
7. **Amplitude:** The maximum value (positive or negative) attained by an alternating quantity in one cycle is called its 'amplitude or peak value or maximum value'. The maximum value of voltage and current is generally denoted by E_m (or V_m) and I_m , respectively.

Values of alternating voltage and current

1. Peak value
2. Average value or Mean value
3. Effective value or rms value

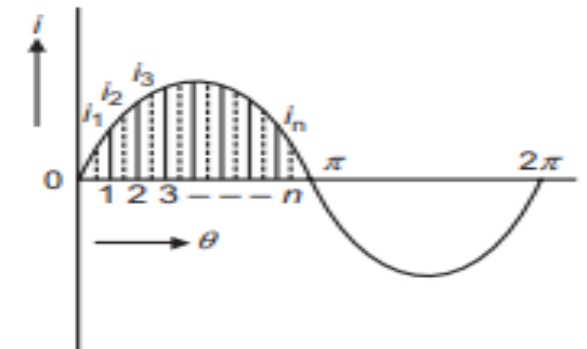
1. Peak value

The maximum value attained by an alternating quantity during one cycle is called 'peak value'. This is also called 'maximum value or crest value or amplitude'. A sinusoidal alternating quantity obtains its maximum value at 90° . The peak of an alternating voltage and current is represented by E_m and I_m . The knowledge of peak value is important in case of testing dielectric strength of insulating materials.

2. Average value or Mean value

The arithmetic average of all the instantaneous values considered an alternating quantity (current or voltage) over one cycle is called average value

$$\begin{aligned} &= \frac{i_1 + i_2 + i_3 + \dots + i_n}{n} \\ &= \frac{\text{Area of alternation}}{\text{Base}} \end{aligned}$$



Average value of sinusoidal current

$$i = I_m \sin \theta$$

Consider an elementary strip of thickness $d\theta$ in the positive half cycle, i be its mid-ordinate. Then,

$$\text{Area of strip} = i d\theta$$

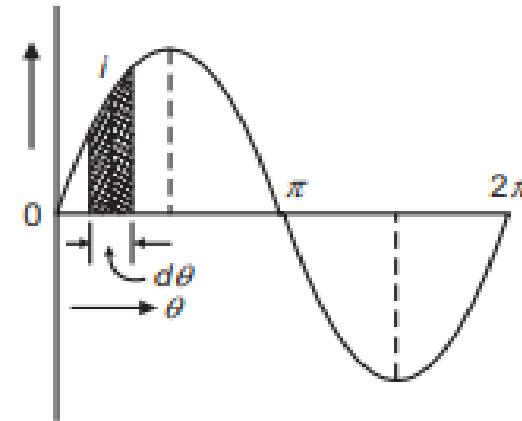
$$\text{Area of half cycle} \int_0^{\pi} i d\theta = \int_0^{\pi} I_m \sin \theta d\theta$$

$$= I_m (-\cos \theta)_0^{\pi} = I_m (-(\cos \pi - \cos 0))$$

$$= I_m [-1(-1-1)] = 2I_m$$

$$\text{Base} = 0 \text{ to } \pi = \pi - 0 = \pi$$

$$\therefore \text{Average value, } I_{av} = \frac{\text{Area of alternation}}{\text{base}} = \frac{2I_m}{\pi} = \frac{I_m}{\pi/2} = 0.637 I_m$$



Current varying sinusoidally

Effective or rms value

The steady current when flows through a resistor of known resistance for a given time produces the same amount of heat as produced by an alternating current when flows through the same resistor for the same time is called effective or rms value of an alternating current

Then, heat produced in the

First interval = $i_1^2 Rt/Jn$ calorie

Second interval = $i_2^2 Rt/Jn$ calorie

Third interval = $i_3^2 Rt/Jn$ calorie

n th interval = $i_n^2 Rt/Jn$ calorie

Total heat produced = $\frac{Rt}{J} \left(\frac{i_1^2 + i_2^2 + i_3^2 + \dots + i_n^2}{n} \right)$ calorie

Since I_{eff} is considered as the effective value of this current.

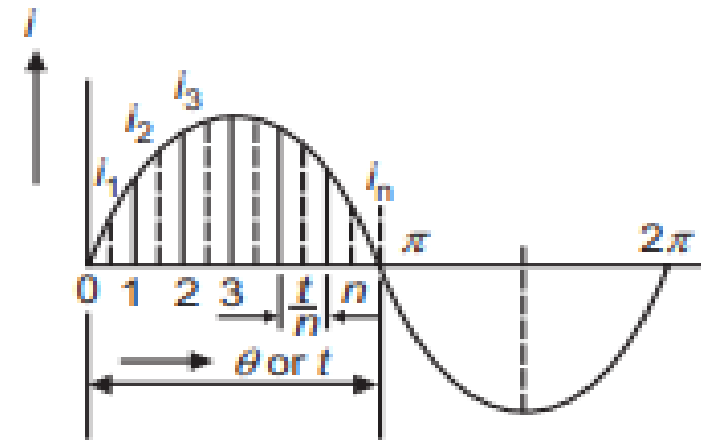
Then, total heat produced by this current = calorie

Equating equation (6.3) and (6.4), we get,

$$\frac{I_{\text{eff}}^2 Rt}{J} = \frac{Rt}{J} \left(\frac{i_1^2 + i_2^2 + i_3^2 + \dots + i_n^2}{n} \right)$$

$$I_{\text{eff}} = \sqrt{\frac{i_1^2 + i_2^2 + i_3^2 + \dots + i_n^2}{n}} = \sqrt{\text{mean of squares of instantaneous values}}$$

I_{eff} = Square root of mean of squares of instantaneous values
= root-mean-square (rms) value



Positive half cycle divided into n equal part

RMS value of sinusoidal current

$$\text{Area of strip} = i^2 d\theta$$

Area of first half cycle of squared wave

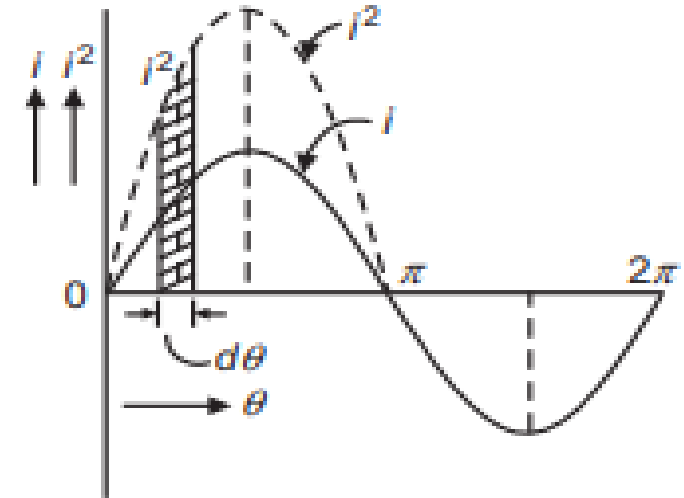
$$\begin{aligned} &= \int_0^{\pi} i^2 d\theta = \int_0^{\pi} (I_m \sin \theta)^2 d\theta \\ &= I_m^2 \int_0^{\pi} \sin^2 \theta d\theta = I_m^2 \int_0^{\pi} \frac{1 - \cos 2\theta}{2} d\theta \\ &= \frac{I_m^2}{2} \int_0^{\pi} (1 - \cos 2\theta) d\theta = \frac{I_m^2}{2} \left(\theta - \frac{\sin 2\theta}{2} \right)_0^{\pi} \\ &= \frac{I_m^2}{2} \left((\pi - 0) - \frac{\sin 2\pi - \sin 0}{2} \right) \\ &= \frac{I_m^2}{2} [(\pi - 0) - (0 - 0)] = \frac{\pi I_m^2}{2} \end{aligned}$$

Base = 0 to $\pi = \pi - 0 = \pi$

Effective or rms value,

$$I_{\text{rms}} = \sqrt{\frac{\text{Area of first half of squared wave}}{\text{Base}}} = \sqrt{\frac{\pi I_m^2}{2\pi}} = \sqrt{\frac{I_m^2}{2}} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$$

Usually, rms value of an AC is simply represented by I instead of I_{rms} . Similarly, rms value of an alternating voltage is represented by E or V .



Squared-wave shape of a sine wave

1. **Form factor:** The ratio of rms value to average value of an alternating quantity is called form factor.

$$\text{Mathematically, form factor} = \frac{I_{\text{rms}}}{I_{\text{av}}} \text{ or } \frac{E_{\text{rms}}}{E_{\text{av}}}$$

For the current varying sinusoidally

$$\text{Form factor} = \frac{I_{\text{rms}}}{I_{\text{av}}} = \frac{I_m / \sqrt{2}}{2I_m / \pi} = \frac{\pi I_m}{2\sqrt{2} I_m} = 1.11$$

2. **Peak factor:** The ratio of maximum value to rms value of an alternating quantity is called peak factor.

$$\text{Mathematically, peak factor} = \frac{I_m}{I_{\text{rms}}} \text{ or } \frac{E_m}{E_{\text{rms}}}$$

For current varying sinusoidally

$$\text{Peak factor} = \frac{I_m}{I_{\text{rms}}} = \frac{I_m}{I_m / \sqrt{2}} = \sqrt{2} = 1.4142$$

An alternating voltage is given by $v = 141.1 \sin 314t$. Find the following:

- Frequency
 - RMS value
 - Average value
 - The instantaneous value of voltage when 't' is 3 msec
 - The time taken for voltage to reach 100 V for the first time after passing through zero value.
- (U.P.T.U. 2006–2007)

Solution:

(i) Given that, $v = 141.14 \sin 314 t$

Comparing above equation with

$$v = V_m \sin \omega t$$

$$\omega = 314 \text{ rad/sec.}$$

$$\therefore \text{Frequency, } f = \frac{314}{2\pi} = \frac{314}{2 \times 3.14} = 50 \text{ Hz (Ans.)}$$

(ii) RMS value of the voltage, $V_{\text{rms}} = \frac{V_m}{\sqrt{2}} = \frac{141.4}{1.414} = 100 \text{ V (Ans.)}$

(iii) Average value, $V_{\text{av}} = \frac{2V_m}{\pi} = \frac{2 \times 141.4}{\pi} = 90 \text{ V (Ans.)}$

(iv) $v = 141.1 \sin 314t$.

$$\text{At } t = 3 \text{ ms} = 3 \times 10^{-3} \text{ s}$$

$$v = 141.4 \sin 314 \times 3 \times 10^{-3} = 114.4 \text{ V (Ans.)}$$

(v) $v = 141.4 \sin 314t$

or $t = \frac{1}{314} \sin^{-1} \left(\frac{100}{141.4} \right) = 2.5 \times 10^{-3} \text{ s} = 2.5 \text{ ms (Ans.)}$

7.2 AC CIRCUIT CONTAINING RESISTANCE ONLY

The circuit containing a pure resistance of $R \Omega$ is shown in Figure 7.1. Let the alternating voltage applied across the circuit be given by the equation;

$$v = V_m \sin \omega t \quad (7.1)$$

Then, the instantaneous value of current flowing through the resistor will be;

$$i = \frac{v}{R} = \frac{V_m}{R} \sin \omega t \quad (7.2)$$

The value of current will be maximum when

$$\omega t = 90^\circ \quad \text{or} \quad \sin \omega t = 1$$

$$I_m = V_m / R$$

Substituting this value in equation (7.2), we get

$$i = I_m \sin \omega t \quad (7.3)$$

7.2.1 Phase Angle

From Equations (7.1) and (7.3), it is clear that there is no phase difference between the applied voltage and the current flowing through pure resistive circuit, that is, phase angle between the voltage and the current is zero. The phasor diagram and wave diagram are shown in Figure 7.2(a) and (b), respectively.

Hence, in an AC circuit containing pure resistance, current is in phase with the voltage.

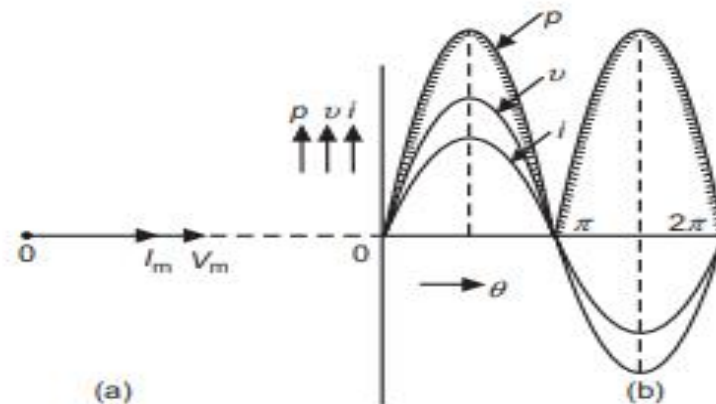


Fig. 7.2 (a) Phasor diagram (b) Wave diagram for voltage, current and power

7.2.2 Power

Instantaneous power, $p = vi = (V_m \sin \omega t)(I_m \sin \omega t)$

$$= \frac{V_m I_m}{2} 2 \sin^2 \omega t$$

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} (1 - \cos 2\omega t) = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} - \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos 2\omega t$$

Average power consumed in the circuit over a complete cycle:

$$P = \text{average of } \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} - \text{average of } \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \omega t$$

or $P = V_{\text{rms}} I_{\text{rms}} - 0 \text{ or } P = VI$

7.3 AC CIRCUIT CONTAINING PURE INDUCTANCE ONLY

The circuit containing a pure inductance of L Henry is shown in Figure 7.3.

Let the alternating voltage applied across the circuit be given by the equation;

$$v = V_m \sin \omega t \quad (7.4)$$

As a result, an AC i flows through the inductance that induces an emf in it, given by the relation;

$$e = -L \frac{di}{dt}$$

This induced emf is equal and opposite to the applied voltage.

$$\therefore v = -e = -\left(-L \frac{di}{dt}\right)$$

or

$$V_m \sin \omega t = L \frac{di}{dt} \quad \text{or} \quad di = \frac{V_m}{L} \sin \omega t \, dt$$

Integrating both sides

$$\int di = \int \frac{V_m}{L} \sin \omega t \, dt \quad \text{or} \quad i = \frac{V_m}{\omega L} (-\cos \omega t)$$

or

$$i = \frac{V_m}{\omega L} \sin (\omega t - \pi/2) = \frac{V_m}{X_L} \sin (\omega t - \pi/2)$$

where $X_L = \omega L$ is the opposition offered to the flow of AC by a pure inductance and it is called inductive reactance.

The value of current will be maximum when $\sin (\omega t - \pi/2) = 1$; i.e., $I_m = \frac{V_m}{X_L}$

$$\therefore i = I_m \sin (\omega t - \pi/2) \quad (7.5)$$

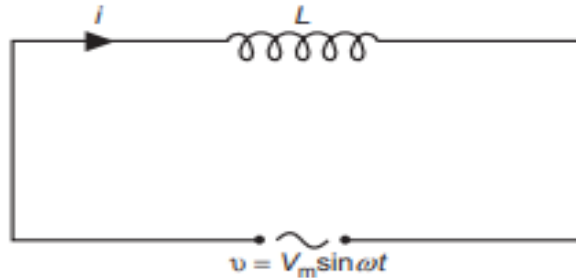


Fig. 7.3 Circuit containing pure inductance only

7.3.1 Phase Angle

From Equations (7.4) and (7.5), it is clear that current flowing through a pure inductive circuit lags behind the applied voltage v by 90° . The phasor diagram is shown in Figure 7.4(a) and 7.4(b), respectively.

Hence, in an AC circuit containing pure inductance, current lags behind the voltage by 90° .

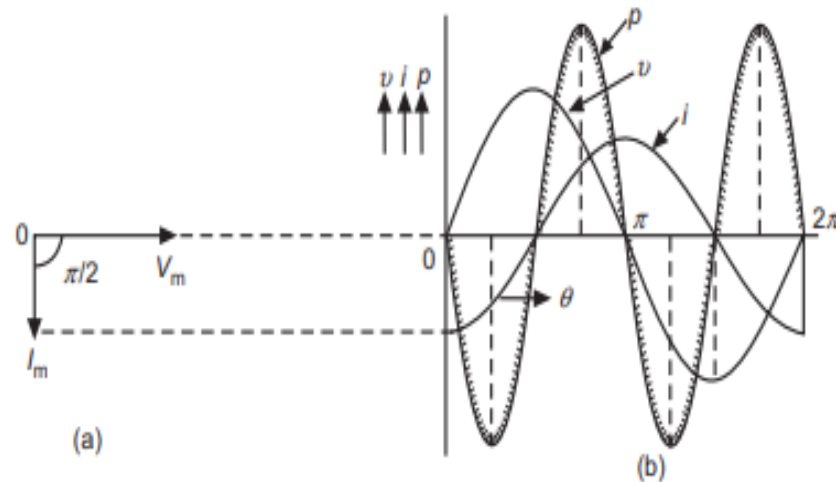


Fig. 7.4 (a) Phasor diagram (b) Wave diagram for voltage, current and power

7.3.2 Power

Instantaneous power, $p = vi = V_m \sin \omega t \times I_m \sin (\omega t - \pi/2)$

$$= V_m I_m \sin \omega t \cos \omega t = \frac{V_m I_m}{2} 2 \sin \omega t \cos \omega t$$

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \sin 2\omega t$$

Average power consumed in the circuit over a complete cycle,

$$P = \text{average} \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \sin 2\omega t = 0$$

Hence, average power consumed in a pure inductive circuit is zero.

7.4 AC CIRCUIT CONTAINING PURE CAPACITOR ONLY

The circuit containing a pure capacitor of capacitance C Farad is shown in Figure 7.5. Let the alternating voltage applied across the circuit be given as

$$v = V_m \sin \omega t \quad (7.6)$$

Charge on the capacitor at any instant,

$$q = Cv$$

Current flowing through the circuit,

$$i = \frac{d}{dt}q = \frac{d}{dt}(Cv)$$

or

$$i = \frac{d}{dt}CV_m \sin \omega t = CV_m \frac{d}{dt} \sin \omega t$$

or

$$i = \omega C V_m \cos \omega t = \frac{V_m}{1/\omega C} \sin(\omega t + \pi/2) = \frac{V_m}{X_C} \sin(\omega t + \pi/2) \quad (7.7)$$

where $X_C = 1/\omega C$ is the opposition offered to the flow of AC by a pure capacitor and is called capacitive reactance.

The value of current will be maximum when $\sin(\omega t + \pi/2) = 1$

i.e.,

$$I_m = V_m/X_C$$

Substituting this value in Equation (7.7), we get

$$i = I_m \sin(\omega t + \pi/2) \quad (7.8)$$

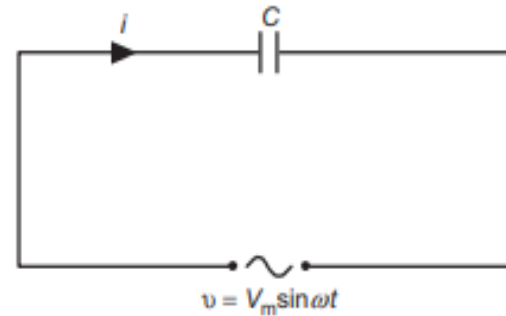


Fig. 7.5 Circuit diagram containing pure capacitor only

7.4.1 Phase Angle

From Equations (7.6) and (7.8), it is clear that the current flowing through pure capacitive circuit leads the applied voltage by 90° . The phasor diagram and wave diagram are shown in Figure 7.6(a) and (b), respectively. Hence, in an AC circuit containing pure capacitance current leads the voltage by 90° .

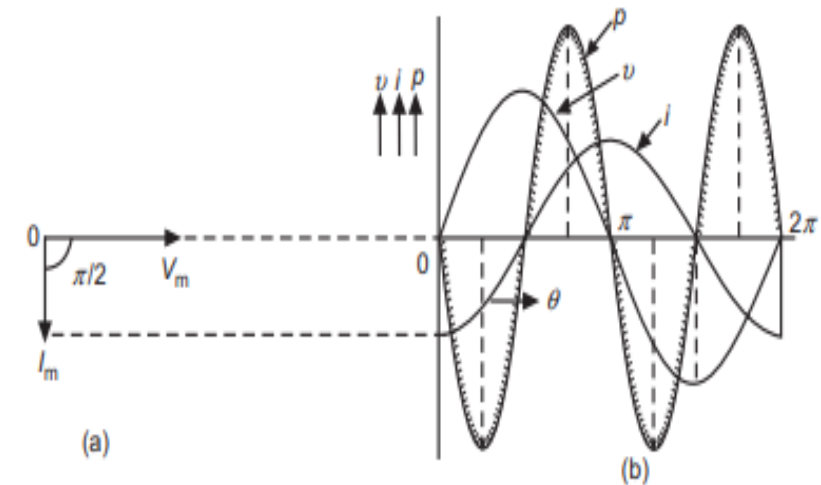


Fig. 7.6 (a) Phasor diagram (b) Wave diagram for voltage, current and power

7.4.2 Power

Instantaneous power, $p = vi = V_m \sin \omega t \times I_m \sin (\omega t + \pi/2)$

$$= V_m I_m \sin \omega t \cos \omega t = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \sin 2 \omega t$$

or average power over a complete cycle,

$$P = 0$$

Hence, average power consumed in a pure capacitive circuit is zero.

A capacitor has a capacitance of $30 \mu\text{F}$. Find its capacitive reactance for frequencies of 25 and 50 Hz. Find in each case the current if the supply voltage is 440 V.

Solution:

Capacitance of the capacitor, $C = 30 \times 10^{-6} \text{ F}$

Supply voltage, $V = 440 \text{ V}$

When supply frequency, $f_1 = 25 \text{ Hz}$

$$\text{Capacitive reactance, } X_{C1} = \frac{1}{\omega_1 C} = \frac{1}{2\pi f_1 C} = \frac{1}{2\pi \times 25 \times 30 \times 10^{-6}} = 212.2 \Omega$$

$$\text{Current in the circuit, } I_1 = \frac{V}{X_{C1}} = \frac{440}{212.2} = 2.073 \text{ A}$$

When supply frequency $f_2 = 50 \text{ Hz}$,

$$\text{capacitive reactance, } X_{C2} = \frac{1}{\omega_2 C} = \frac{1}{2\pi f_2 C} = \frac{1}{2\pi \times 50 \times 30 \times 10^{-6}} = 106.1 \Omega$$

$$\text{Current in the circuit, } I_2 = \frac{V}{X_{C2}} = \frac{440}{106.1} = 4.146 \text{ A}$$

AC series circuits

1. R–L series circuit
2. R–C series circuit
3. R–L–C series circuit

R–L series circuit

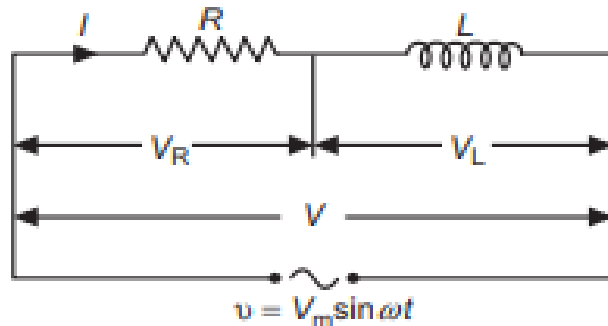


Fig. 7.7 Circuit containing resistance and inductance in series

Now, $V_R = IR$ and $V_L = IX_L$ (where $X_L = 2\pi fL$)

In right-angled triangle OAB,

$$V = \sqrt{(V_R)^2 + (V_L)^2} = \sqrt{(IR)^2 + (IX_L)^2} = I\sqrt{R^2 + X_L^2}$$

or

$$I = \frac{V}{\sqrt{R^2 + X_L^2}} = \frac{V}{Z}$$

where $Z = \sqrt{R^2 + X_L^2}$ is the total opposition offered to the flow of AC by an R–L series circuit and is called impedance of the circuit. It is measured in ohms.

$$\text{From phasor diagram, } \tan \phi = \frac{V_L}{V_R} = \frac{IX_L}{IR} = \frac{X_L}{R} \text{ or } \phi = \tan^{-1} X_L/R$$

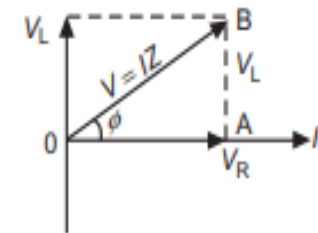


Fig. 7.8 Phasor diagram

Power

If the alternating voltage applied across the circuit is given by the equation.

$$v = V_m \sin \omega t$$

Then,

$$i = I_m \sin (\omega t - \phi)$$

$\therefore$ Instantaneous power, $p = vi$

$$= V_m \sin \omega t I_m \sin(\omega t - \phi) = \frac{V_m I_m}{2} 2 \sin \omega t \sin(\omega t - \phi)$$

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} [\cos \phi - \cos(2\omega t - \phi)]$$

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi - \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(2\omega t - \phi)$$

Average power consumed in the circuit over a complete cycle,

$$P = \text{average of } \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi - \text{average of } \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(2\omega t - \phi)$$

or

$$P = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi - \text{zero} = V_{\text{rms}} I_{\text{rms}} \cos \phi = VI \cos \phi$$

where $\cos \phi$ is called power factor of the circuit.

$$\text{From phasor diagram, } \cos \phi = \frac{V_R}{V} = \frac{IR}{IZ} = \frac{R}{Z}$$

Therefore, power factor is defined as the cosine of the angle between the voltage and the current in an AC circuit. It may also be defined as the ratio of resistance to impedance of an AC circuit.

$$\text{Alternatively, power, } P = VI \cos \phi = IZ \cdot I \cdot \frac{R}{Z} = I^2 R$$

This shows that power is actually consumed in resistance only; inductance does not consume any power.

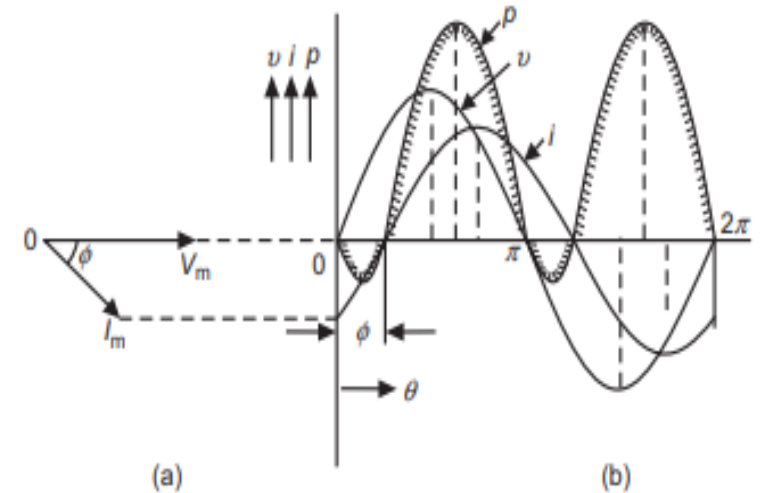


Fig. 7.9 (a) Phasor diagram (b) Wave diagram for voltage, current and power

The simplified phasor diagram of R-L series circuit is shown in Figure 7.10. When each side of this phasor diagram is divided by a common factor I , we get another right-angled triangle, as shown

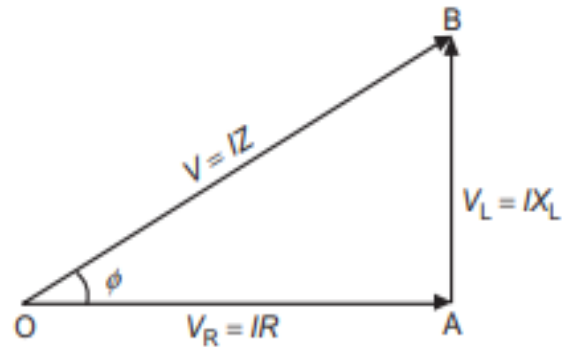


Fig. 7.10 Phasor diagram for R-L series circuit

in Figure 7.11, whose sides represent R , X_L , and Z . Such a triangle is known as impedance triangle.

Therefore, a right-angled triangle whose base represents circuit resistance, perpendicular represents circuit reactance, and hypotenuse represents circuit impedance is called an impedance triangle. The concept of impedance triangle is useful since it enables us to calculate:

- (i) the impedance of the circuit,

$$Z = \sqrt{R^2 + X^2}$$

- (ii) the power factor of the circuit,

$$\cos \phi = R/Z$$

- (iii) phase angle, $\phi = \tan^{-1} X_L/R$

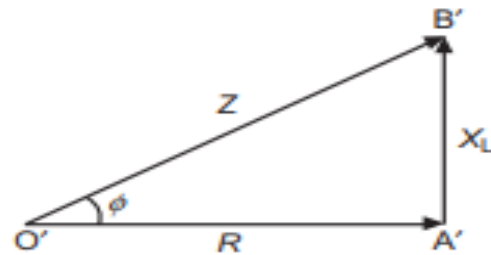


Fig. 7.11 Impedance triangle

true power = voltage $\times$ current in phase with voltage

reactive power = voltage $\times$ current 90° out of phase with voltage.

The phasor diagram for an inductive circuit is shown in Figure 7.12, where current I lags behind the voltage V by an angle ϕ° . Current I can be resolved into two rectangular components, that is, (i) $I \cos \phi$, which is in phase with voltage V and (ii) $I \sin \phi$, which is 90° out of phase with voltage V .

$\therefore$ True power, $P = V \times I \cos \phi = VI \cos \phi \text{ W}$

Reactive power, $P_r = V \times I \sin \phi = VI \sin \phi \text{ VAR}$

Apparent power, $P_a = V \times I = VI \text{ VA}$

The bigger units of true power, reactive power, and apparent power are kW (or MW), kVAR (or MVAR), and kVA (or MVA), respectively.

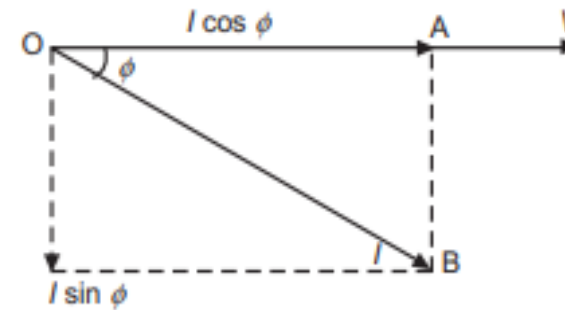


Fig. 7.12 Phasor diagram representing active, reactive and apparent current

When each component of current, in Figure 7.12, is multiplied by voltage V , a power triangle is obtained, as shown in Figure 7.13. This right-angled triangle indicates the relation among true power, reactive power, and apparent power.

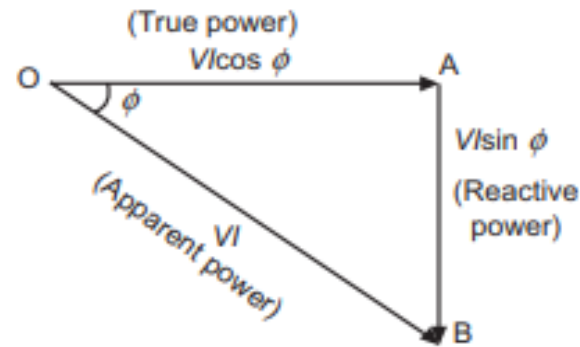


Fig. 7.13 Power triangle

In the abovementioned discussion, the following points are worth noting:

1. When an active component of current is multiplied with circuit voltage, it results in active or true power. It is this power that produces torque in motors, heat in heaters, light in lamps, etc. Further, watt-meter indicates this power.
2. When the reactive component of current is multiplied with circuit voltage, it results in reactive power. It is this power that merely flows back and forth without doing any work. This power determines the power factor of the circuit.
3. When the circuit current is multiplied with circuit voltage, it results in apparent power. It is so called because it appears that product of voltage and current is power. However, in AC circuits (except pure resistive circuit), there is usually phase difference between voltage and current so that VI does not give real power. To avoid confusion, it is measured in volt-ampere.
4. From power triangle shown in Figure 7.13, the power factor may also be determined by taking ratio of true power to apparent power, that is, power factor, $\cos \phi = \text{true power} / \text{apparent power}$.

Importance of Power Factor

$$P = VI \cos \phi \text{ or } I = \frac{P}{V \cos \phi}$$

From this relation, it is clear that for fixed power at constant voltage, the current drawn by the circuit increases with a decrease in pf. Therefore, at low pf, AC circuits draw more current from their mains and results in the following disadvantages:

1. **Greater conductor size:** At low pf, the conductors are to carry more current for the same power, and therefore, they require large area of cross-section.
2. **Poor efficiency:** At low power factors, the conductors have to carry larger current that increases copper losses (I^2R) and results in poor efficiency.
3. **Larger voltage drop:** At low power factors, the conductors have to carry large current that increases voltage drop (IR) in the system and results in poor regulation.
4. **Larger kVA rating of equipment:** The kVA rating of electrical machines and equipment connected in the power system such as alternators, transformers, and switch gears will be more at low power factors since it is inversely proportional to power factor (i.e., $\text{kVA} = \text{kW}/\cos \phi$).

To improve the power factor of an AC circuit, a capacitor is connected across the circuit, that is, parallel to the circuit.

Q-factor of a coil

Reciprocal of power factor of a coil is known as its Q -factor. It is also called quality factor or figure of merit of the coil.

Mathematically,
$$Q\text{-factor} = \frac{1}{\text{p.f.}} = \frac{1}{\cos \phi} = \frac{Z}{R}$$

If the value of R is very small in comparison to its inductive reactance X_L , then

$$Q\text{-factor} = \frac{X_L}{R} = \frac{\omega L}{R}$$

Further,
$$Q = 2\pi \times \frac{\text{Maximum energy stored}}{\text{Energy dissipated per cycle}}$$

A non-inductive resistance of $10\ \Omega$ is connected in series with an inductive coil across 200 V , 50 Hz AC supply. The current drawn by the series combination is 10 A . The resistance of the coil is $2\ \Omega$. Determine (i) inductance of the coil, (ii) power factor, and (iii) voltage across the coil. **(U.P.T.U. 2005–06)**

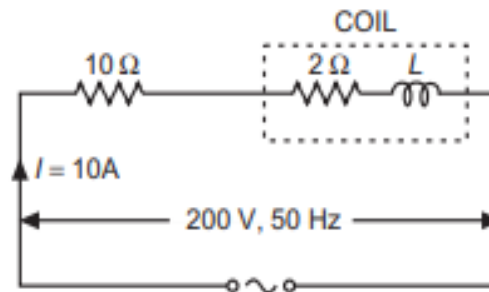


Fig. 7.17 Circuit as per data

Solution:

The circuit as per data is shown in Figure 7.17

Total impedance of the circuit, $Z = \frac{V}{I} = \frac{200}{10} = 20\ \Omega$

Total resistance of the circuit, $R = 10 + 2 = 12\ \Omega$

Inductive reactance of the coil, $X_L = \sqrt{Z^2 - R^2} = \sqrt{(20)^2 - (12)^2} = 16\ \Omega$

or $2\pi fL = 16$ or $L = \frac{16}{100\pi} = 50.93\text{ mH}$

$\therefore$ Inductance of the coil, $L = 50.93\text{ mH}$

(ii) Power factor of the circuit, $\cos \phi = \frac{R}{Z} = \frac{12}{20} = 0.6\text{ lag}$

(iii) Impedance of the coil, $Z_c = \sqrt{R_c^2 + X_L^2} = \sqrt{(2)^2 + (16)^2} = 16.124\ \Omega$

Voltage across the coil, $V_c = IZ_c = 10 \times 16.124 = 161.24\text{ V}$

Power factor of the coil, $\cos \phi_c = \frac{R_c}{Z_c} = \frac{2}{16.124} = 0.124\text{ lagging}$

R-C series circuit

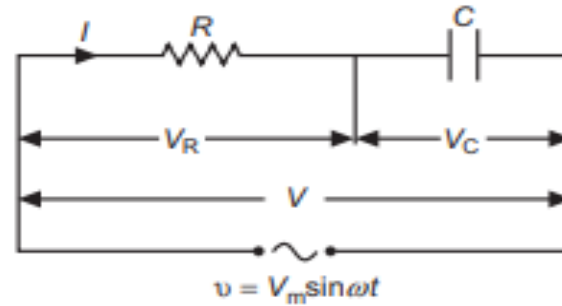


Fig. 7.21 Circuit containing resistance and capacitance in series

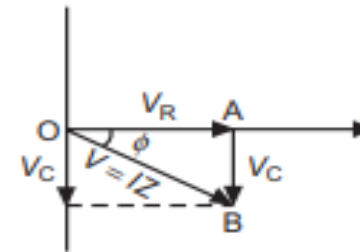


Fig. 7.22 Phasor diagram

Now, $V_R = IR$ and $V_C = IX_C$ (where $X_C = 1/2 \pi fC$)

In right-angled triangle OAB

$$V = \sqrt{(V_R)^2 + (V_C)^2} = \sqrt{(IR)^2 + (IX_C)^2} = \sqrt{R^2 + X_C^2}$$

or

$$I = \frac{V}{\sqrt{R^2 + X_C^2}} = \frac{V}{Z}$$

where $Z = \sqrt{R^2 + X_C^2}$ is the total opposition offered to the flow of AC by an R-C series circuit and is called impedance of the circuit. It is measured in ohm.

$$\tan \phi = \frac{V_C}{V_R} = \frac{IX_C}{IR} = \frac{X_C}{R} \quad \text{or} \quad \phi = \tan^{-1} X_C/R$$

Power

If the alternating voltage applied across the circuit is given by the equation:

$$v = V_m \sin \omega t$$

Then,

$$i = I_m \sin (\omega t + \phi)$$

$\therefore$ Instantaneous power,

$$\begin{aligned} p &= vi = V_m \sin \omega t I_m \sin (\omega t + \phi) = \frac{V_m I_m}{2} 2 \sin (\omega t + \phi) \sin \omega t \\ &= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} [\cos \phi - \cos (2\omega t + \phi)] = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi + \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos (2\omega t + \phi) \end{aligned}$$

Average power consumed in the circuit over a complete cycle,

$$P = \text{average of } \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi - \text{average of } \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos (2\omega t + \phi)$$

or
$$P = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \cos \phi = V_{rms} I_{rms} \cos \phi = VI \cos \phi$$

where $\cos \phi$ is called power factor of the circuit.

From phasor diagram,

$$\cos \phi = \frac{V_R}{V} = \frac{IR}{IZ} = \frac{R}{Z} \text{ same as in R-L series circuit}$$

Alternatively, power

$$P = VI \cos \phi = IZ \cdot I \cdot \frac{R}{Z} = I^2 R$$

This shows that power is actually consumed in resistance only; capacitor does not consume any power.

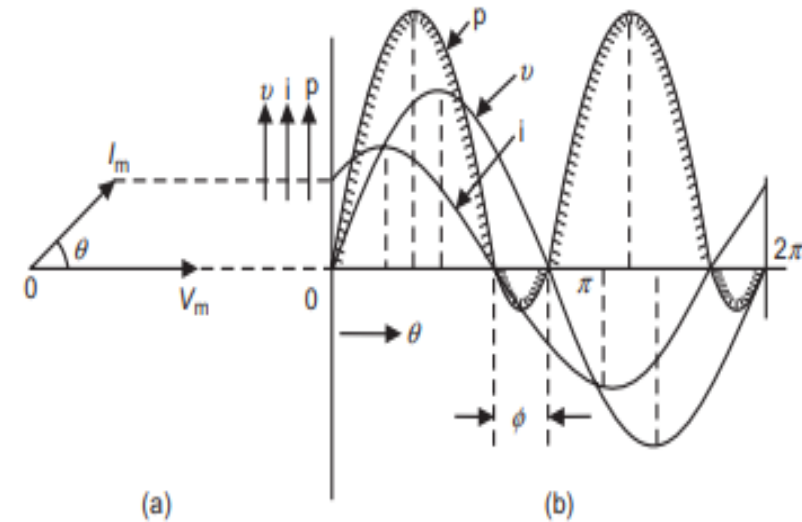


Fig. 7.23 (a) Phasor diagram (b) Wave diagram for voltage, current and power

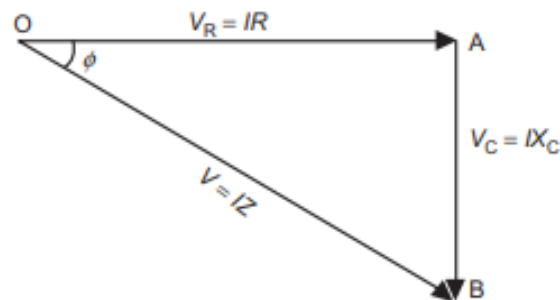


Fig. 7.24 Phasor diagram for R-C series circuit

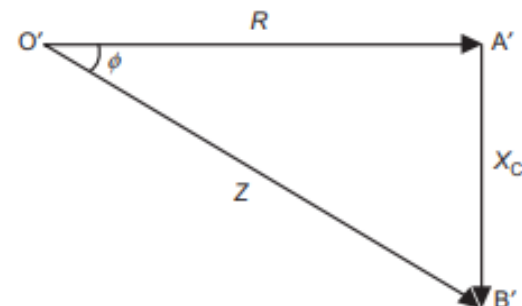
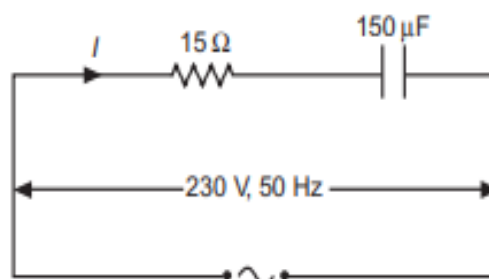


Fig. 7.25 Impedance triangle

A resistance of $15\ \Omega$ and capacitor of $150\ \mu\text{F}$ capacitance are connected in series across a $230\ \text{V}$, $50\ \text{Hz}$ supply. Calculate (i) impedance of the circuit, (ii) current, (iii) power factor and phase angle, and (iv) power consumed in the circuit.



Impedance, $Z = \sqrt{R^2 + X_C^2}$

where $X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 150 \times 10^{-6}}$
 $= 21.22\ \Omega$;
 $R = 15\ \Omega$

(i) $\therefore Z = \sqrt{(15)^2 + (21.22)^2} = 25.987\ \Omega$

(ii) Current, $I = \frac{V}{Z} = \frac{230}{25.987} = 8.85\ \text{A}$

(iii) Power factor, $\cos \phi = \frac{R}{Z} = \frac{15}{25.987} = 0.577$ leading

Phase angle, $\phi = \cos^{-1} 0.577 = 54.75^\circ$

(iv) Power, $P = VI \cos \phi = 230 \times 8.85 \times 0.577 = 1,174.9\ \text{W}$

R-L-C series circuit

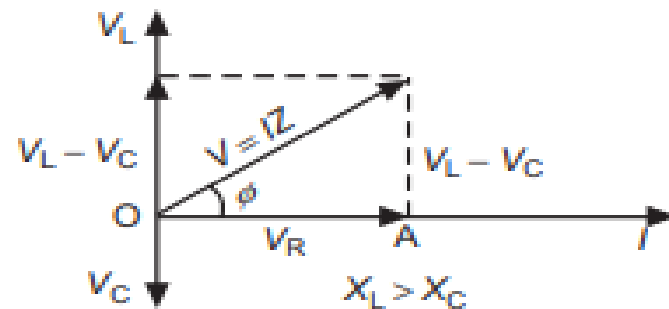
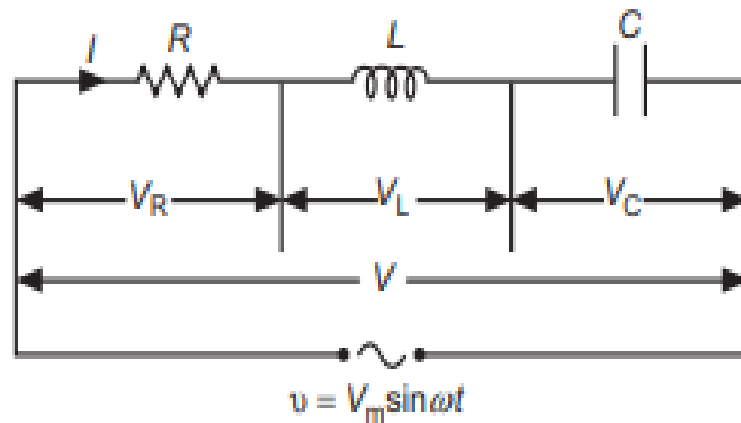


Fig. 7.29 Phasor diagram

When a resulting current I (rms value) flows through the circuit, the voltage across each component will be

$V_R = IR$, that is, voltage across R in phase with I ;

$V_L = IX_L$, that is, voltage across L leads I by 90° ;

$V_C = IX_C$, that is, voltage across C lags I by 90° ;

The phasor diagram is shown in Figure 7.29, where current is taken as the reference phasor. Since voltage across inductance V_L leads the current vector I by 90° and voltage across capacitance V_C lags the current vector I by 90° , they act opposite to each other. If $V_L > V_C$, in effect, the circuit behaves as an inductive circuit; however, when $V_L < V_C$, the circuit behaves as a capacitive circuit. Here, the phasor diagram is drawn for an inductive circuit (i.e., when $V_L > V_C$).

$$V = \sqrt{(V_R)^2 + (V_L - V_C)^2} = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

or

$$V = I\sqrt{R^2 + (X_L - X_C)^2}$$

or

$$I = \frac{V}{\sqrt{(R)^2 + (X_L - X_C)^2}} = \frac{V}{Z}$$

where $Z = \sqrt{R^2 + (X_L - X_C)^2}$ is the total opposition offered to the flow of AC by an R-L-C series circuit and is called impedance of the circuit.

Power

$$\text{From phasor diagram: } \tan \phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R}$$

$$\text{or } \phi = \tan^{-1} \frac{X_L - X_C}{R}$$

Average power, $P = VI \cos \phi = I^2 R$

$$\text{Power factor, } \cos \phi = \frac{V_R}{V} = \frac{R}{Z}$$

The alternating voltage applied across the circuit is given by

$$v = V_m \sin \omega t,$$

Therefore, the circuit current is represented by the equation as per the constants or parameters, as explained in the following three cases of R–L–C series circuit:

1. When $X_L > X_C$, the phase angle ϕ is positive. In effect, the circuit behaves as an R–L series circuit. The circuit current lags behind the applied voltage and pf is lagging. The current is given by the equation.

$$i = I_m \sin (\omega t - \phi)$$

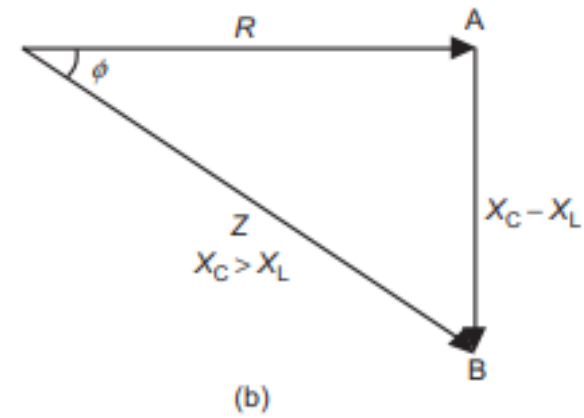
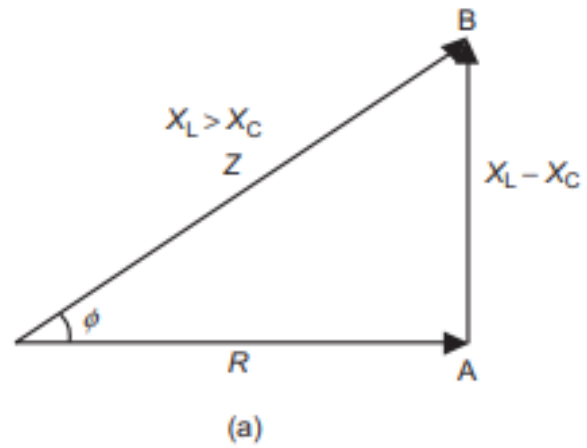
2. When $X_L < X_C$, the phase angle ϕ is negative. In effect, the circuit behaves as an R–C series circuit. The circuit current leads the applied voltage and pf is leading. The current is given by the equation.

$$i = I_m \sin (\omega t + \phi)$$

3. When $X_L = X_C$, the phase angle ϕ is zero. In effect, the circuit behaves like a pure resistive circuit. The circuit current is in phase with applied voltage and pf is unity. The current is given by the equation.

$$i = I_m \sin \omega t$$

Impedance Triangle



SERIES RESONANCE

In an R-L-C series circuit, when circuit current is in phase with the applied voltage, the circuit is said to be in series resonance. This condition is obtained in an R-L-C circuit shown in Figure 7.31,

when $X_L = X_C$ (or $X_L - X_C = 0$)

At resonance, $X_L - X_C = 0$ or $X_L = X_C$

Impedance, $Z_r = \sqrt{R^2 + (X_L - X_C)^2} = R$

Current, $I_r = \frac{V}{Z_r} = \frac{V}{R}$

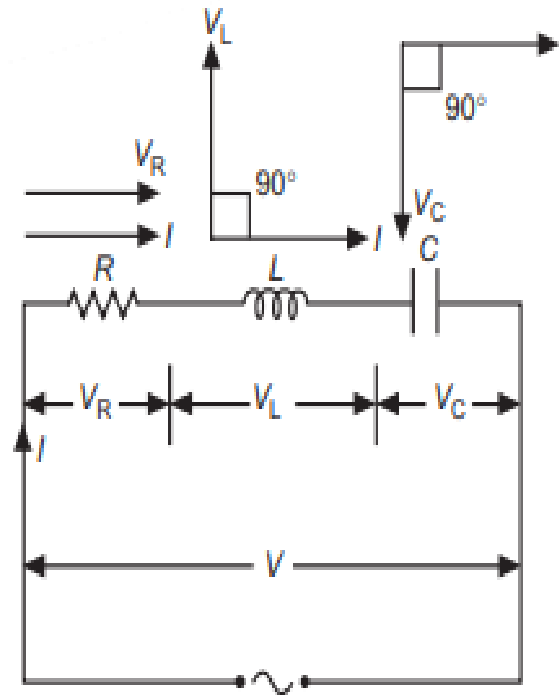


Fig. 7.31 R-L-C series circuit

Resonant Frequency

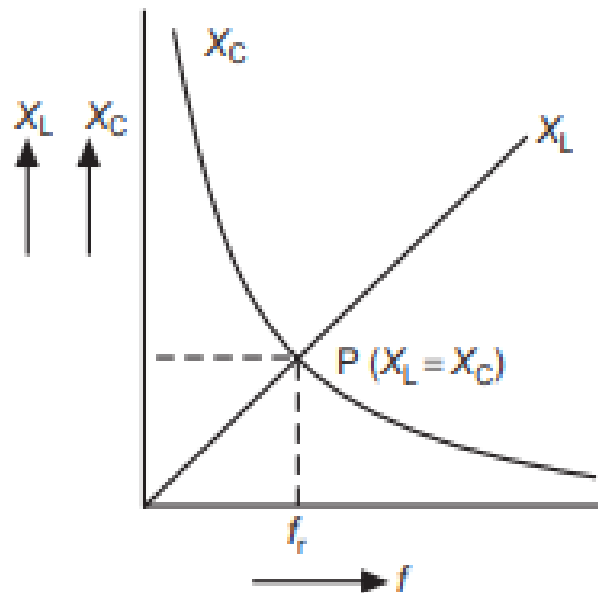


Fig. 7.32 Phasor diagram for series resonant circuit

The value of $X_L (= 2\pi f L)$ and $X_C (= 1/2\pi f C)$ can be changed by changing the supply frequency. When frequency increases, the value of X_L increases, whereas the value of X_C decreases, and vice versa. Therefore, to obtain series resonance, the frequency is adjusted to f_r so that $X_L = X_C$, the condition at point P shown in Figure 7.32.

At series resonance, $X_L = X_C$

$$2\pi f_r L = \frac{1}{2\pi f_r C} \quad \text{or} \quad f_r = \frac{1}{2\pi\sqrt{LC}}$$

where f_r is the resonant frequency in Hz when L and C are measured in Henry and Farad, respectively.

Effects of Series Resonance

The following are the main effects of series resonance:

- (i) At resonance $X_L = X_C$, and therefore, the impedance of the circuit is minimum and is reduced to the resistance of the circuit only, that is,

$$Z_r = R$$

- (ii) Since, impedance is minimum, the circuit current is maximum at resonance, that is

$$I_r = V/Z_r = V/R$$

- (iii) Power taken by the circuit is maximum, as I_r is maximum,

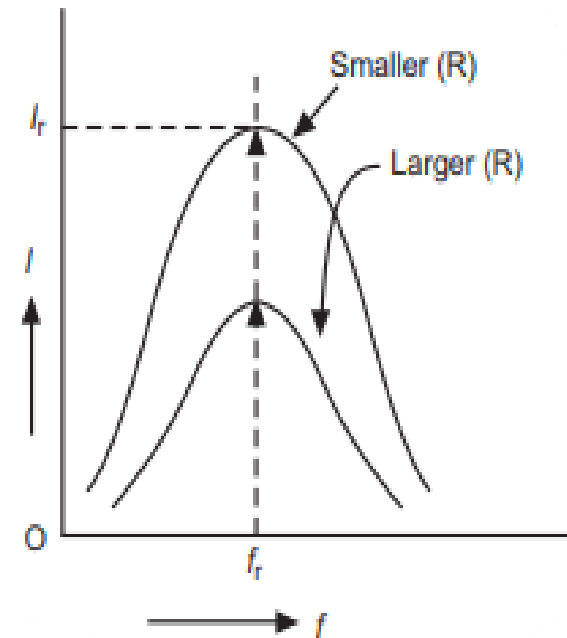
$$P_r = I_r^2 R$$

- (iv) As the current drawn by the circuit, at resonance, is very large (maximum), the voltage drop across L (i.e., $V_L = IX_L = I \times 2\pi f_r L$) and (i.e., $V_C = IX_C = I \times 1/2\pi f_r C$) are also very large. In power system, at resonance, the excessive voltage built up across the inductive and capacitive components (such as circuit breakers and reactors) may cause damage. Therefore, series resonance should be avoided in power system. However, in some of the electronic devices (such as antenna circuit of radio and TV receiver and tuning circuits), the principle of series resonance is used to increase the signal voltage and current at a desired frequency (f_r). Since a series resonant circuit has the capability to draw heavy current and power from the mains, it is often regarded as acceptor circuit.

RESONANCE CURVE

It may be noted that current reaches its maximum value at the resonant frequency (f_r), falling off rapidly on either side of that point. It is because when the value of frequency is lower than resonance frequency, $X_c > X_L$ and when the value of frequency is higher than f_r , $X_c < X_L$. In both the cases, impedance of the circuit increases ($Z > Z_r$) and the value of current decreases.

Note that resistance of the circuit also plays its own role. The smaller the resistance, the greater is the current at resonance.



Bandwidth

The range of frequency over which circuit current is equal to or more than 70.7% of maximum value (i.e., I_r , current at resonance) is known as the bandwidth of a series resonant circuit.

Figure 7.34 shows a resonance curve of a typical R-L-C circuit where the circuit current is equal to or greater than 70.7% of maximum current (i.e., $I_r = V/R$) between frequency range f_1 to f_2 .

Bandwidth, $BW = f_2 - f_1$

Here, the frequency f_1 is called the lower cut-off frequency and the frequency f_2 is called the upper cut-off frequency. The bandwidth represents the frequency range at which the circuit offers low impedance to circuit current.

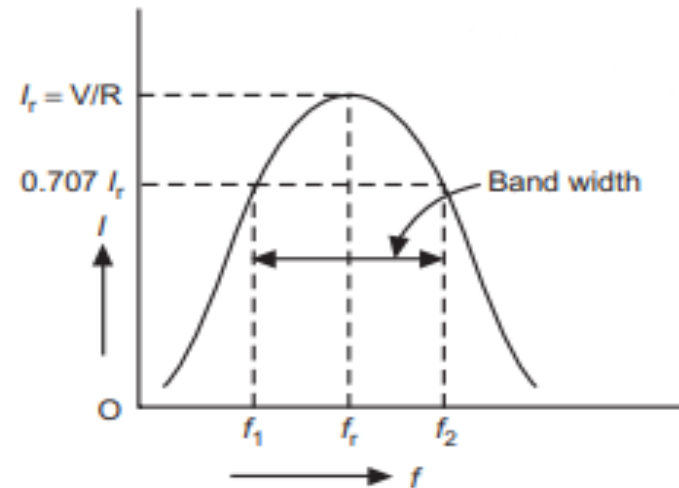
The following points may be noted here:

1. If the resonant frequency is not located at the centre of upper and lower cut-off frequency, then

$$F_r = \sqrt{f_1 f_2}$$

2. When the resonant frequency is located sufficiently near to the centre of the two cut-off frequencies and Q of the circuit is ≥ 10 , then

$$f_1 = f_r - \frac{BW}{2} \quad \text{and} \quad f_2 = f_r + \frac{BW}{2}$$



Q-FACTOR OF SERIES RESONANT CIRCUIT

We have seen that at series resonance, the circuit draws the largest current from the mains, this produces a heavy voltage across L or C . The factor by which the potential difference across L or C increases to that of the applied voltage is called the Q -factor of the series resonant circuit.

$$Q\text{-factor} = \frac{\text{Voltage across } L \text{ or } C}{\text{Applied voltage}} = \frac{I_r X_L}{I_R} = \frac{X_L}{R} = \frac{\omega_r L}{R}$$

where

$$\omega_r = 2\pi f_r = 2\pi \frac{1}{2\pi\sqrt{LC}} = \frac{1}{\sqrt{LC}}$$

$$Q\text{-factor} = \frac{L}{R} \times \frac{1}{\sqrt{LC}} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

The value of Q -factor depends entirely upon the design of coil (i.e., R - L , which is a part of R - L - C circuit).

A coil resistance $10\ \Omega$ and inductance $0.14\ \text{H}$ is connected in series with a capacitor of $150\ \mu\text{F}$ across a $200\ \text{V}$, $50\ \text{Hz}$ supply. Calculate (i) inductive reactance, (ii) capacitive reactance, (iii) impedance, (iv) current, and (v) voltage across coil and capacitor. (U.P.T.U. Tut.)

Solution:

The circuit is shown in Figure 7.35.

Let us consider $R = 10\ \Omega$; $L = 0.14\ \text{H}$; $C = 150\ \mu\text{F} = 150 \times 10^{-6}\ \text{F}$; $V = 200\ \text{V}$; $f = 50\ \text{Hz}$

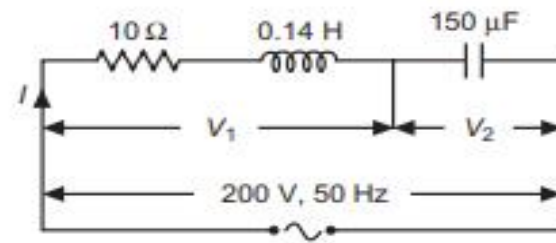


Fig. 7.35 Circuit as per data

(i) Inductive reactance,

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.14 = 44\ \Omega$$

(ii) Capacitive reactance,

$$X_C = \frac{1}{2\pi fC}$$

$$= \frac{1}{2\pi \times 50 \times 150 \times 10^{-6}} = 21.22\ \Omega$$

(iii) Impedance, $Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{10^2 + (44 - 21.22)^2}$

$$= 24.88\ \Omega$$

(iv) Current, $I = \frac{V}{Z} = \frac{200}{24.88} = 8.04\ \text{A}$

(v) Voltage across coil, $V_1 = IZ_{\text{COIL}} = I\sqrt{R^2 + X_L^2}$

$$= 8.04\sqrt{10^2 + 44^2} = 362.8\ \text{V}$$

Voltage across capacitor, $V_2 = IX_C = 8.04 \times 21.22 = 170.6\ \text{V}$



THANKS